

1(a)	<i>independent variable:</i> concentration / molarity, of sucrose solution ; <i>dependent variable:</i> number of, plasmolysed / non-plasmolysed, <u>cells</u> ;	2
1(b)(i)	(weigh / AW) 85.5 g sucrose and make up to 250 cm ³ with, (distilled / deionised), water ;	1
1(b)(ii)	range of at least 5 concentrations from 0 to 1.0 mol dm ⁻³ sucrose at (equidistant) intervals ; shows proportions of (distilled / deionised) water and 1.0 mol dm ⁻³ sucrose to make up a volume of 20 cm ³ of at least two different concentrations ;	2
1(b)(iii)	idea that pigment makes it much easier to, see cells / observe changes in vacuole size / judge plasmolysis ;	1

Question	Answer	Marks
1(b)(iv)	<i>any eight from:</i> 1 ref. to method of making epidermal strips (from onion leaves) ; e.g. tweezers / fingers / scalpel / needle 2 ref. to cutting epidermal strips to a standard size / ref. to standardising sample size of epidermal cells ; 3 ref. to at least one, strip / sample of epidermal cells, into different concentration solutions (in container or on a slide) ; 4 ref. to making sure, strips / epidermal cells, are completely immersed in the solutions ; 5 ref. to leaving for same period of time ; 6 ref. to mounting on a slide in (appropriate) sucrose solution (with coverslip) ; 7 ref. to using a (light) microscope to count cells ; 8 ref. to counting, large number, of cells / ref. to making two counts from same strip / ref. to counting cells in two fields of view ; 9 ref. to counting / recording number of plasmolysed cells and number of non-plasmolysed / turgid cells AW ; 10 ref. to minimum of two replicates or counting three different strips from the same solution and taking a mean ; 11 ref. to calculation of percentage by multiplying proportion of plasmolysed cells by 100 ; 12 ref. to low / medium risk ;	8
1(b)(v)	plot a graph of concentration of sucrose solution against percentage plasmolysis ; draw a line at 50% plasmolysis and read off the concentration ; OR find concentration above and below / closest to those that give a percentage plasmolysis of 50% ; make solutions in this range and repeat experiment to find the concentration giving 50% plasmolysis ;	2

Question	Answer	Marks
1(c)	1 red onion (epidermis cells) (−0.85) and antirrhinum (epidermis cells) (−1.07) cells will both lose water and because the water potential / MPa is, less negative / higher/ greater, than the solution or the solution water potential, is more negative / lower / less, than the cells ; 2 rhubarb (epidermis cells) (−1.22) would gain water and because the water potential / MPa is, more negative / lower / lesser than the solution or the solution water potential is, less negative / higher / greater than the cells ;	2

Question	Answer	Marks
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